

MONTCELO, AR, USA

WMO: 720175

Lat: 33.636N

Lon: 91.756W

Elev: 290

StdP: 14.54

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53919

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.5	26.3	8.9	8.8	29.3	13.9	11.3	31.0	18.6	58.7	16.9	57.3	3.7	40	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	98.4	77.4	95.6	76.7	93.3	76.6	80.5	91.4	79.6	90.6	78.6	89.2	6.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.7	145.8	85.5	76.6	140.2	84.8	75.4	134.5	83.9	44.2	91.1	43.2	90.7	42.2	89.6	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	13.6	12.1	16.1	101.5	4.3	3.8	13.0	104.2	10.4	106.4	8.0	108.5	4.9	111.3
			14.5	82.6	4.0	2.1	11.6	84.2	9.2	85.4	7.0	86.6	4.1	88.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.5	44.6	47.7	56.1	64.9	72.5	79.7	82.5	82.5	76.3	65.2	54.4	46.7
	DBStd	15.87	10.92	10.46	9.68	7.93	6.31	4.23	4.21	4.49	6.31	8.54	9.17	9.92
	HDD50	698	239	156	49	5	0	0	0	0	0	3	58	188
	HDD65	2560	635	491	304	97	16	0	0	0	4	106	334	573
	CDD50	5997	73	91	239	451	699	891	1008	1006	789	475	191	84
	CDD65	2385	4	6	28	94	250	441	543	541	343	113	17	5
	CDH74	24578	5	27	200	765	2179	4599	6213	6109	3353	1022	95	11
	CDH80	11126	0	1	28	162	765	2123	3054	3133	1515	333	12	0
Wind	WSAvg	5.1	6.0	6.1	6.3	6.3	5.2	4.4	4.2	3.8	3.8	4.4	5.0	5.5
Precipitation	PrecAvg	53.8	4.9	4.7	5.3	5.4	5.4	3.7	3.2	3.0	3.0	4.5	4.4	5.8
	PrecMax	76.6	13.3	11.1	11.8	19.3	13.1	11.4	7.5	11.1	9.1	20.0	11.5	9.6
	PrecMin	33.5	0.6	1.0	1.3	1.5	1.3	0.8	0.7	0.5	0.2	0.2	1.9	1.1
	PrecStd	12.9	2.6	3.1	2.6	3.7	2.9	2.7	2.2	2.3	2.2	4.3	2.6	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.3	77.8	83.3	88.0	92.1	98.0	101.3	101.9	98.9	91.1	81.6	75.3
		MCWB	63.6	64.0	66.3	70.2	74.4	75.2	76.3	77.4	75.4	70.7	67.5	67.6
	2%	DB	70.3	73.0	79.5	83.9	89.6	94.9	97.3	98.9	94.8	87.7	76.8	70.8
		MCWB	62.2	63.0	64.8	68.7	73.5	75.9	78.0	78.5	74.1	71.9	65.7	64.6
	5%	DB	66.3	69.2	75.3	81.5	87.5	92.5	94.6	96.0	91.8	83.7	73.0	67.2
		MCWB	60.1	61.4	62.7	67.1	72.6	75.9	77.9	77.5	74.3	69.5	63.8	61.5
	10%	DB	62.8	65.2	71.9	79.0	84.4	90.3	92.3	93.1	88.8	80.5	69.9	63.4
		MCWB	56.6	58.7	61.0	65.8	71.6	75.6	77.8	77.0	73.7	67.6	62.4	58.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.1	68.0	69.2	74.0	77.8	80.4	82.0	82.2	79.0	76.8	71.3	69.2
		MCDB	71.5	73.7	78.5	82.8	86.9	89.7	92.2	93.8	90.2	85.4	76.5	73.4
	2%	WB	64.7	65.6	67.2	71.6	76.0	79.1	80.6	80.5	77.8	74.7	68.5	66.2
		MCDB	68.6	70.6	75.7	79.7	84.8	88.9	91.1	92.2	88.4	82.8	74.0	69.9
	5%	WB	61.9	62.8	65.4	70.1	74.8	78.0	79.8	79.4	76.8	72.6	66.2	63.3
		MCDB	65.5	67.7	73.1	78.0	83.4	87.9	90.5	91.2	86.7	79.8	71.5	66.0
	10%	WB	57.2	59.7	63.3	68.5	73.4	77.0	78.8	78.4	75.6	70.4	63.5	59.0
		MCDB	62.0	65.1	69.9	75.9	82.0	86.7	89.2	89.5	84.5	77.1	68.0	62.8
Mean Daily Temperature Range	5% DB	MDBR	18.3	18.6	20.3	21.1	19.6	18.9	18.2	19.8	21.2	21.9	20.4	18.3
		MCDBR	21.5	22.8	23.1	22.6	21.4	21.6	20.6	22.7	24.0	24.4	23.1	20.7
		MCWBR	15.1	14.9	12.1	10.5	8.7	7.7	6.7	7.2	8.0	11.1	13.7	15.1
	5% WB	MCDBR	18.8	19.2	20.5	18.8	18.6	18.2	18.1	19.9	19.3	20.0	18.7	18.7
		MCWBR	15.8	14.6	12.4	10.5	8.9	7.7	7.1	7.5	7.3	11.1	13.5	15.9
Clear-Sky Solar Irradiance	taub	0.320	0.334	0.361	0.401	0.436	0.475	0.489	0.486	0.422	0.361	0.339	0.318	
	taud	2.522	2.477	2.419	2.300	2.260	2.175	2.169	2.168	2.330	2.476	2.493	2.537	
	Ebn at Noon	284	291	290	282	272	260	255	254	267	278	274	279	
	Edh at Noon	26	31	35	41	43	47	47	47	38	30	27	25	
All-Sky Solar Radiation	RadAvg	813	955	1352	1689	1868	2060	1986	1835	1579	1269	911	718	
	RadStd	74	160	118	112	131	147	113	148	134	167	102	78	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page